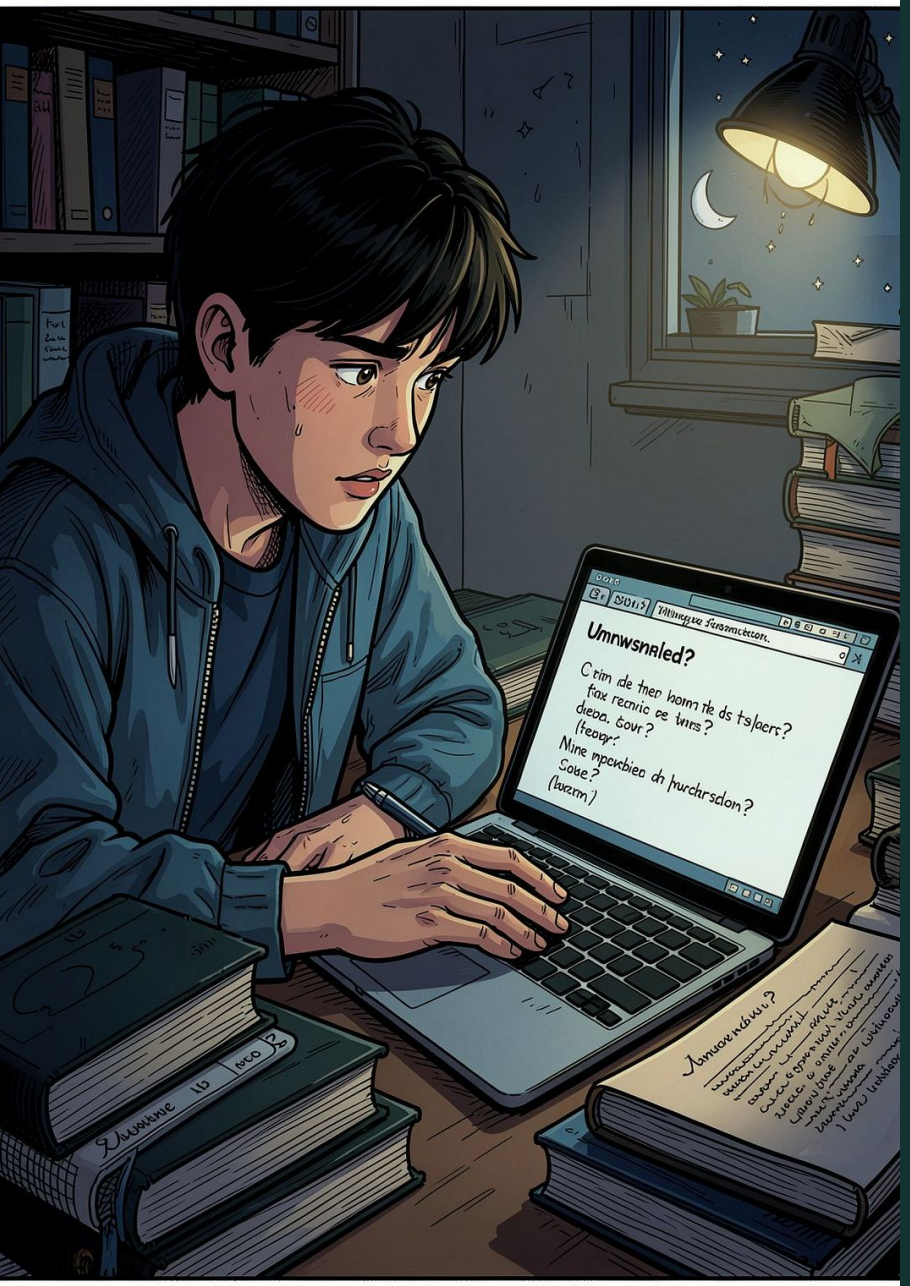


AI Guru — AI-Powered Personalized EdTech Tutor

Learn smarter. Practice better. Grow faster. AI Guru combines adaptive tutoring, retrieval-augmented generation (RAG), active practice, and analytics to deliver on-demand, reliable learning support for students and professionals.



Problem: What Students Really Need

Unlimited doubt support

Students expect instant, continuous help across topics and timezones.

Reliable, grounded answers

Accuracy matters — answers must cite sources and avoid hallucinations.

Personalization & revision

Adaptive practice and revision schedules tuned to each learner's gaps.

Progress tracking

Transparent metrics and actionable insights for learners and coaches.

Solution Snapshot: AI Guru Capabilities

AI Guru delivers a multi-modal tutoring platform: interactive tutor chat, RAG-powered PDF chat, formative quizzes, concise notes, automated code review, personalized roadmap planning, natural-voice tutoring, and learner analytics — all orchestrated by agents.



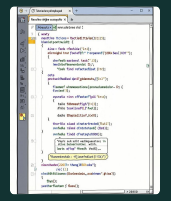
Quizzes & Practice

Adaptive question pools, instant feedback, and mastery tracking.



Notes & Summaries

Auto-generated, skimmable notes linked to source material for revision.



Code Review

Automated review, remediation suggestions, and test-based validation.

Tech Stack — Built for speed, scale, and safety

Core frontend and platform:

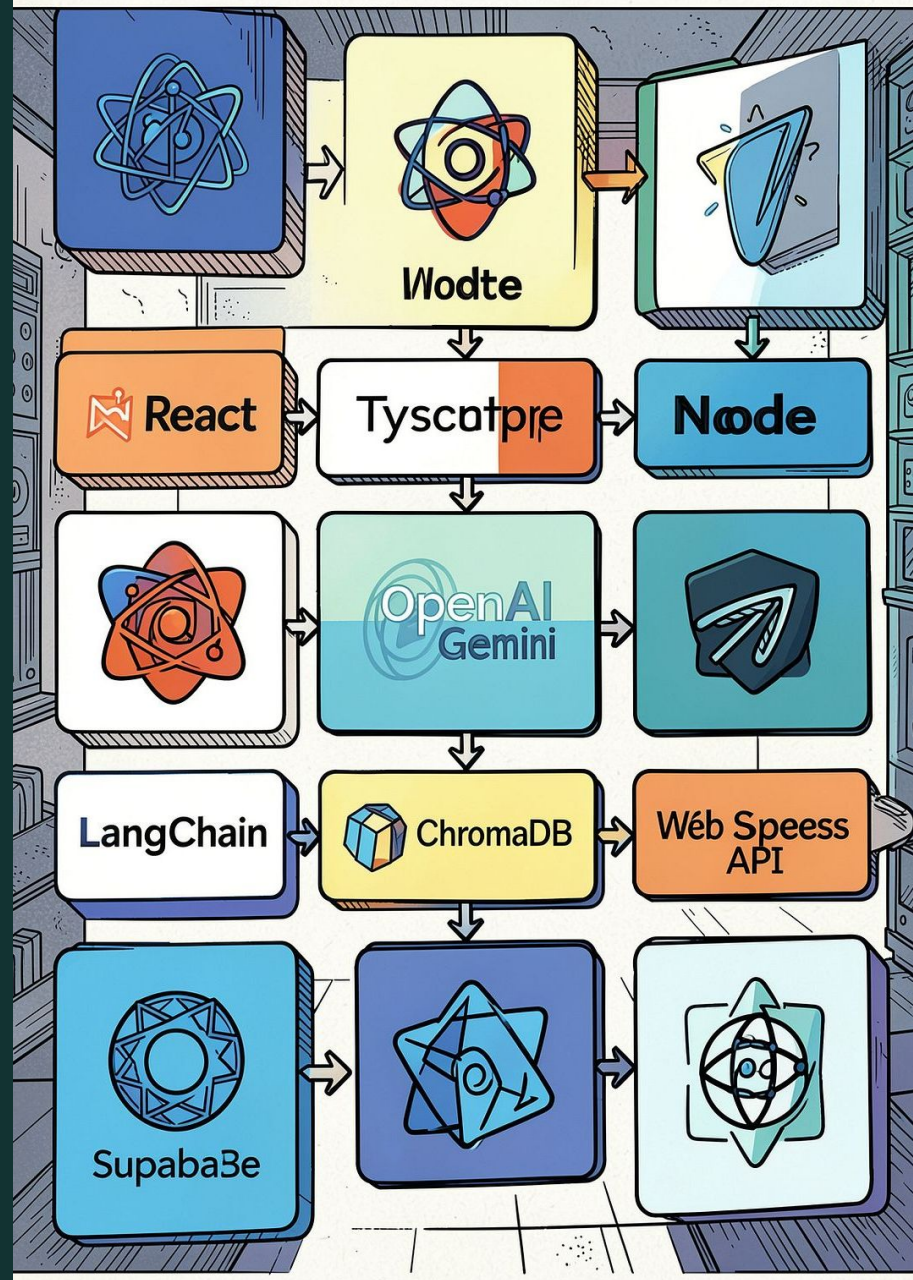
React, TypeScript, Vite for a fast, modular client experience; Node/Express for backend routing and lightweight APIs.

Datastores & infra: Supabase for user and session data; ChromaDB for vector storage and retrieval; hosted object storage for uploads.

AI & orchestration:

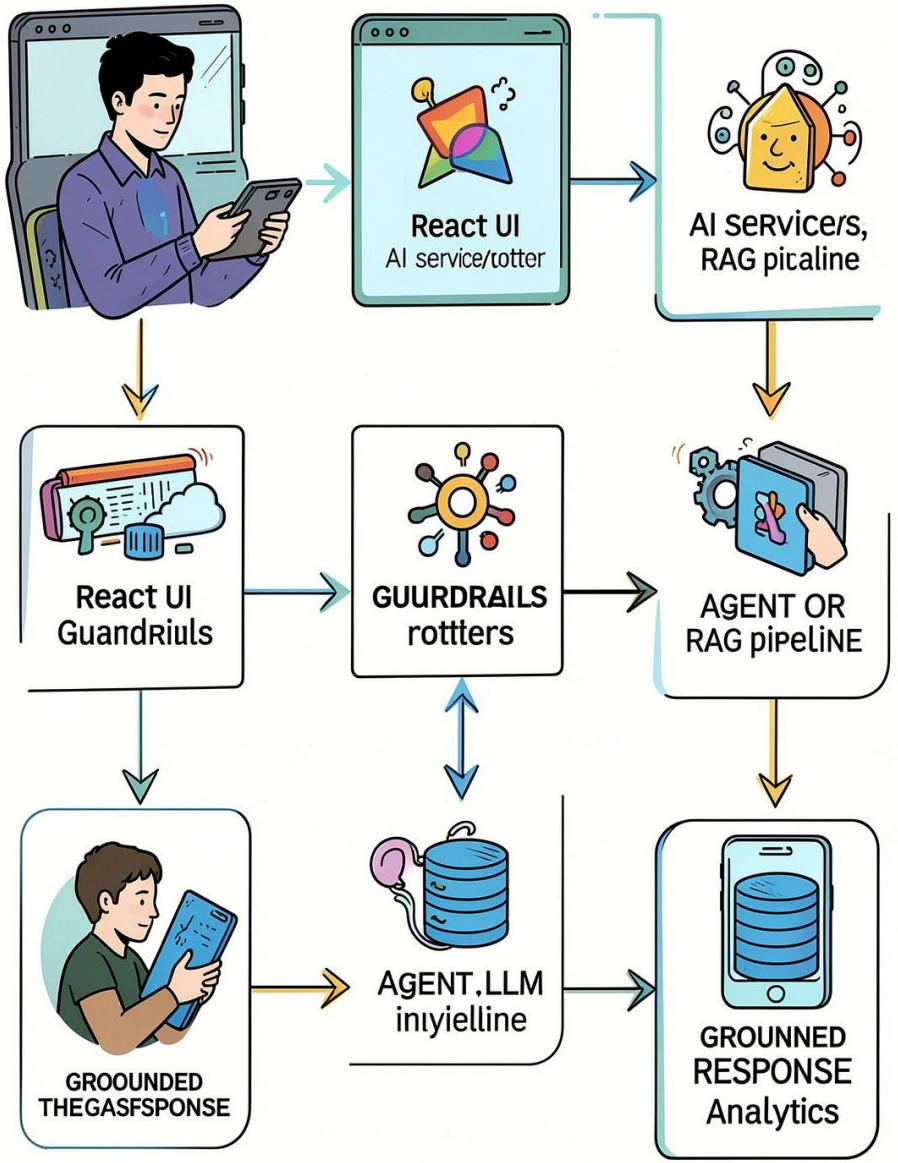
OpenAI GPT and Gemini models for instruction-following and generation; LangChain for agent flows and pipelines; Recharts for analytics visualization.

UX & interactivity: Web Speech API for voice tutoring; monitoring and observability integrated at every layer.



System Architecture (High Level)

Flow: User interacts with the React UI → requests sent to AI service/router → guardrails validate input and select path → Agent pipelines (tutor, planner, quiz) or RAG pipeline invoked → LLM generates responses grounded with retrieved context → response recorded to analytics for monitoring and continuous improvement.



RAG Pipeline: Reliable, Source-Backed Answers

01

1. Ingest & Extract

PDF upload → OCR / text extraction with layout preservation.

02

2. Chunking & Embedding

Semantic chunking, embedding via model, stored in ChromaDB.

03

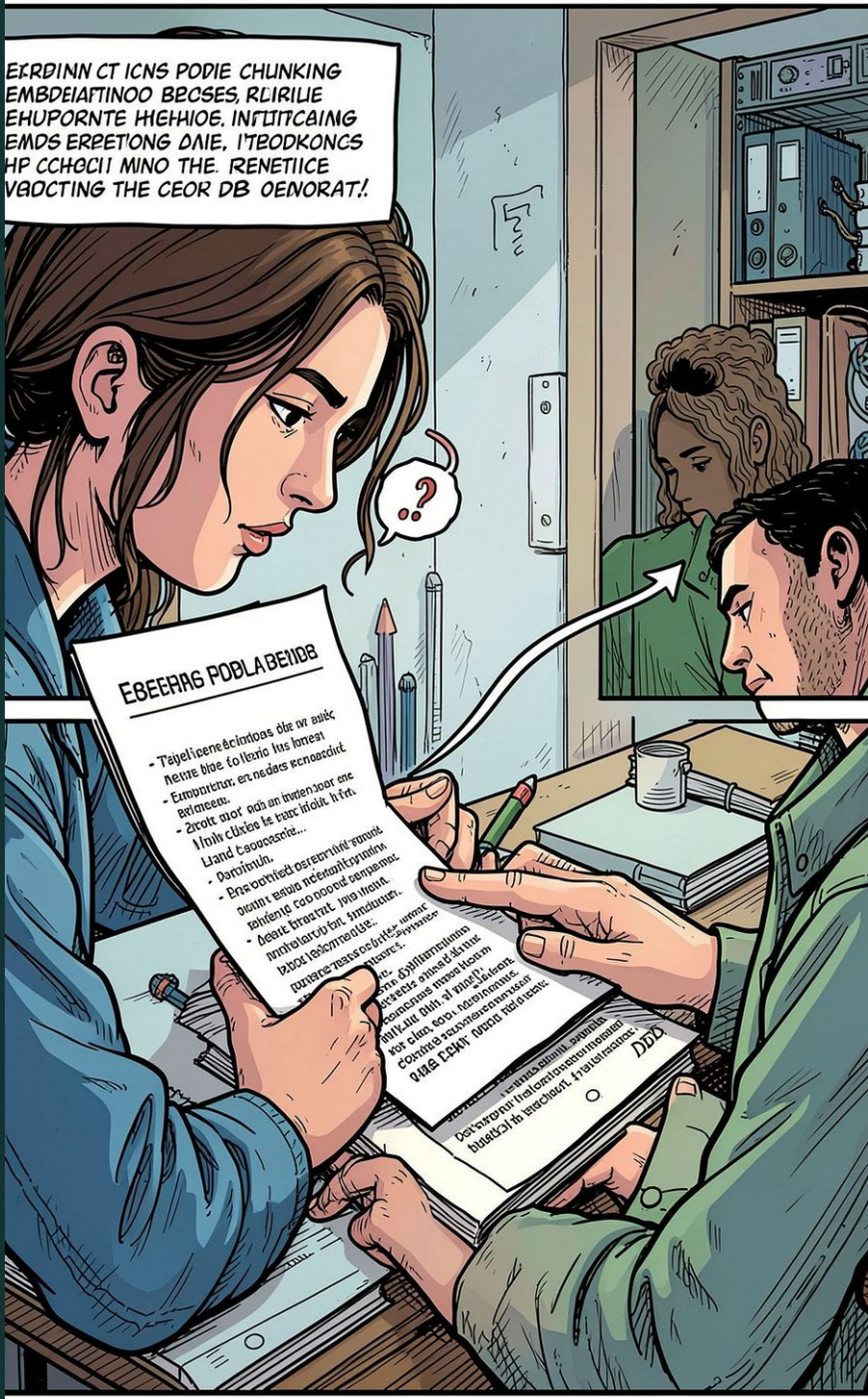
3. Retrieve & Ground

Retriever surfaces high-relevance chunks for the LLM; generation constrained to those chunks.

04

4. Cite & Serve

Responses include citations and source offsets; unknown queries trigger graceful refusal.



AI Agents — Specialized, Coordinated Intelligences



Tutor Agent

Handles conversational teaching, step-by-step explanations, and scaffolding.



Quiz Agent

Generates adaptive questions, grades answers, and updates mastery models.



Evaluator Agent

Evaluates student work against rubrics, flags likely hallucinations, and scores submissions.



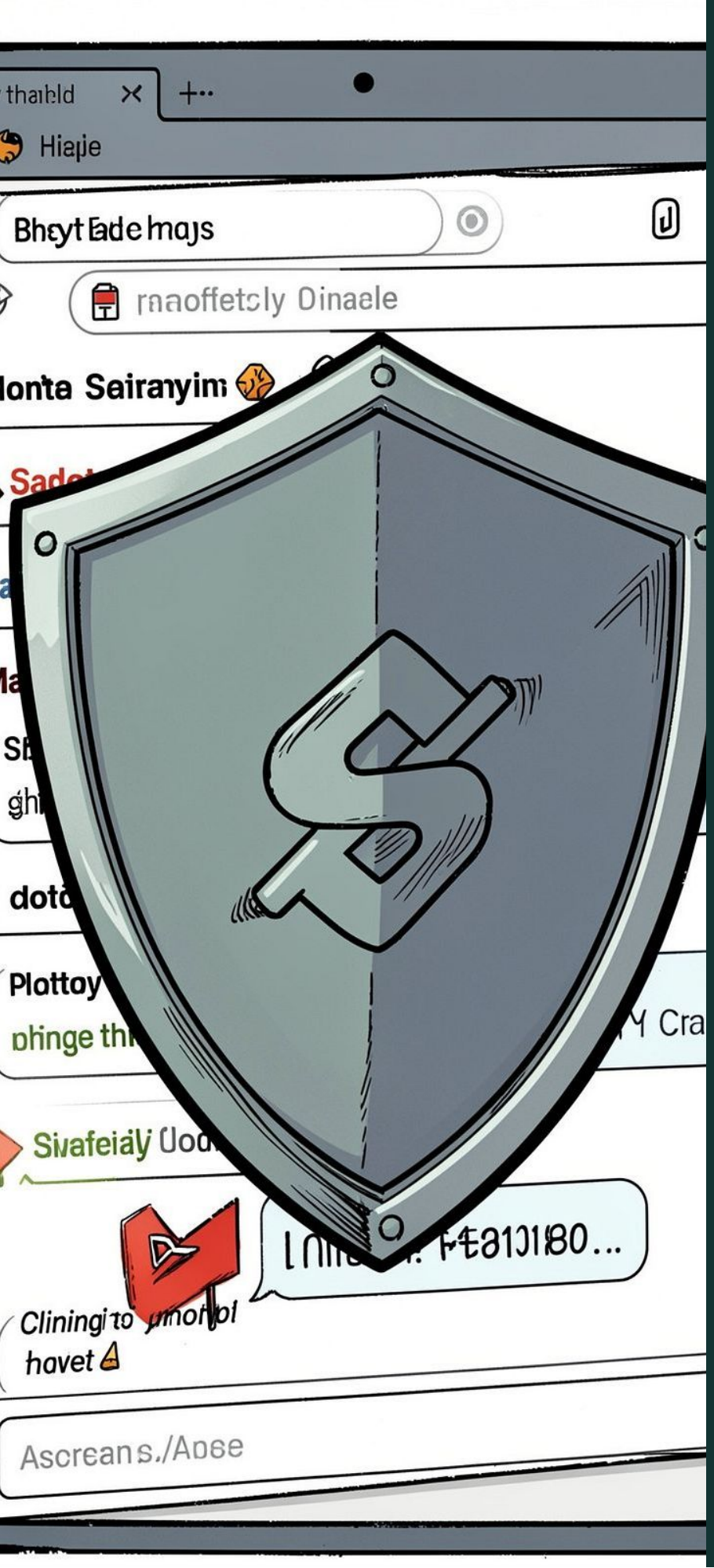
Planner Agent

Builds personalized roadmaps and study schedules based on goals and performance.



Motivation Agent

Delivers nudges, micro-rewards, and motivational coaching to sustain engagement.



Safety & Academic Integrity

1

Prompt injection detection

Input sanitization and policy checks stop malicious prompts before they reach models.

2

Academic integrity checks

Plagiarism detection, answer provenance, and timing heuristics to flag coached submissions.

3

Grounding refusal

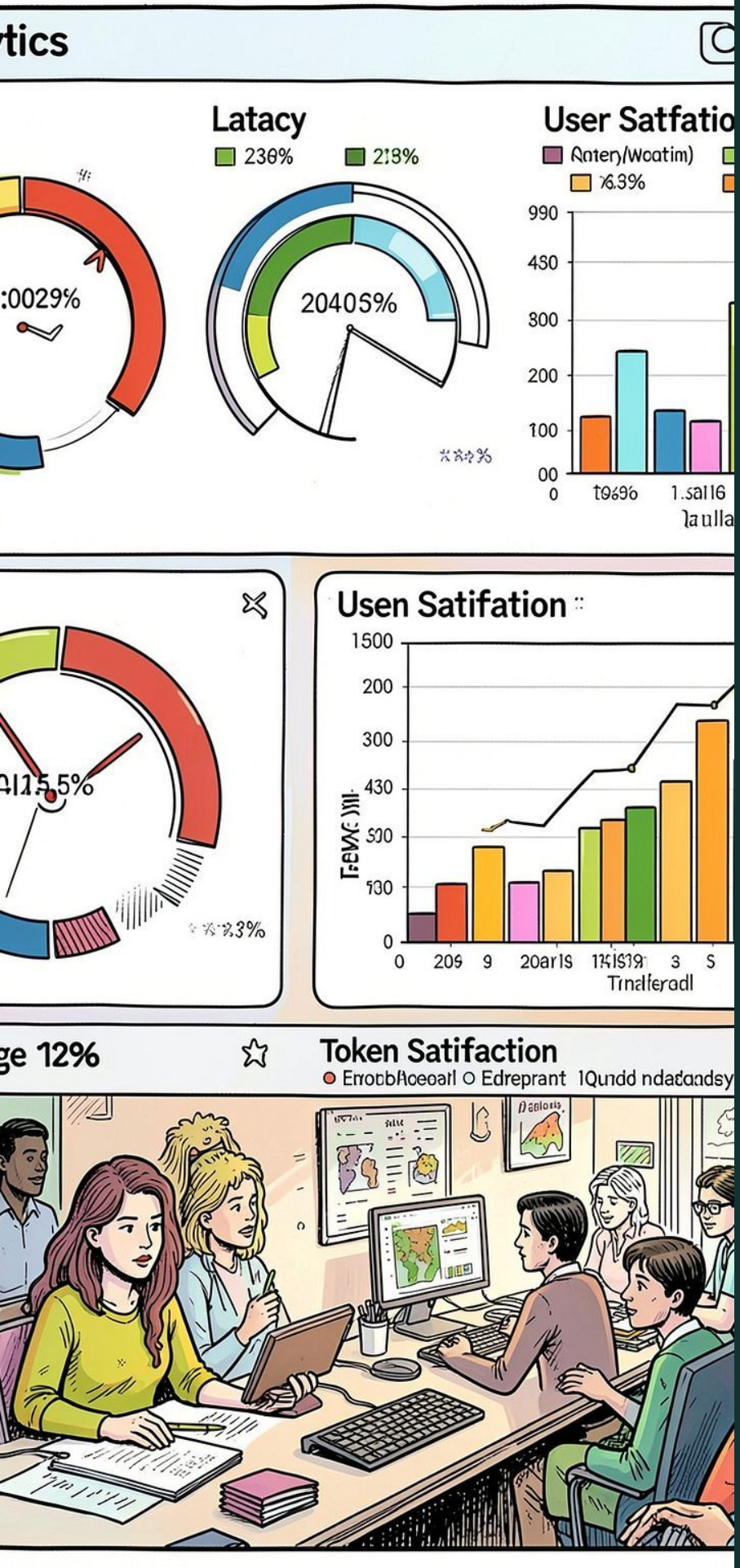
LLMs are constrained to RAG sources; the service refuses to answer when evidence is insufficient.

4

Operational guardrails

Input validation, rate limits, and anomaly detection protect platform stability.

Safety is baked into the pipeline — detection, prevention, and transparent refusal improve trust and compliance.



Evaluation Metrics & Success Criteria

95%

Accuracy target

Goal: ⩾95% factual correctness on grounded responses (RAG-backed).

>90%

Retrieval precision

High-precision retrieval ensures relevant context and lowers hallucination risk.

<2s

Latency

Target response latency under 2 seconds for chat experiences (excluding long RAG searches).

80%

Quiz satisfaction

User-rated usefulness of practice items; drives retention and learning gains.

Additional KPIs: hallucination rate, token usage & cost per session, curriculum coverage, improvement in mastery over time, and instructor adoption rates.

Demo Highlights & Roadmap

1 — Today — Prototype

Interactive tutor UI, basic RAG demo, quiz flow, and analytics mockups.

2 — Q3 — Integrations

Supabase auth/data, ChromaDB vector store, robust eval pipelines, and real API connectors.

3 — Q4 — Scale & Deploy

Production deployment, monitoring, rate-limit policies, and pilot with partner schools.

4 — Future — Growth

Multi-lingual support, expanded curriculum agents, LMS integrations, and enterprise offering.

Demo agenda: dashboard walkthrough, live tutor chat with RAG PDF citation, adaptive quiz demo, automated code review example, and analytics deep-dive. Next steps: finalize APIs, secure Supabase & ChromaDB integration, pilot deployments.

